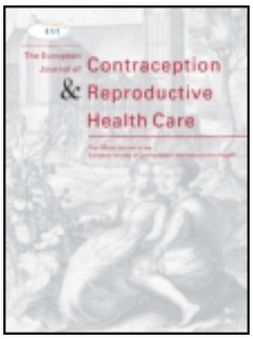

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EPIDEMIOLOGICAL STUDY

Use of oral contraceptives in three European countries: a population-based multi-database study

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ABSTRACT

Objectives The aim of the study was to assess the prevalence of oral contraceptive (OC) use, user characteristics and prescribing patterns by accessing health care databases of three European countries.

Methods A retrospective study was performed from 2009 to 2010 in three general practice (GP) databases from the Netherlands, UK and Italy and in one database of linked pharmacy and hospitalisation data in the Netherlands. The presence of selected chronic conditions and diagnoses of diseases associated with OC use were assessed, as were switches, discontinuations and types of OC used during the study period.

Results Among 2.16 million women aged 15 to 49 years, 16.0% were using an OC on 1 January 2010. The prevalence ranged from 19.7% in a Dutch database to 2.6% in the Italian database. During 2009 and 2010, mainly second-generation progestogens were prescribed in the Netherlands (79.4% and 78.3% of users), both second- (57.9%) and third-generation progestogens (43.6%) were prescribed in the UK, and mainly third-generation progestogens in Italy (61.8%). Most switches were to third- or fourth-generation pills. The prevalence of chronic diseases tended to be higher among OC users, and the proportions of women with a history of disease associated with OC use tended to be lower than among non-users.

Conclusions Second-generation OCs were most frequently prescribed in the Netherlands. In the UK, and even more so in Italy, many women used third- or fourth-generation OCs. Preparation switches were mainly to third- or fourth-generation OCs. Among OC users, a somewhat higher prevalence of chronic diseases was observed; however, information bias cannot be ruled out.

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Database study; descriptive study; Europe; oral contraceptives; prescribing pattern; prevalence

Introduction

Oral contraceptives (OCs) are among the most effective and widely used pharmacological options available for contraception.[1] Since the introduction of OCs more than 50 years ago, several side effects have been associated with their use, in particular venous events (deep vein thrombosis, pulmonary embolism) and arterial events (stroke, myocardial infarction), as well as cancer.[2,3] The risks of these adverse effects are relatively small; therefore, randomised clinical trials have limited power to detect their associations with OC use. To study the occurrence of these adverse effects and prevailing risk factors requires large datasets obtained from daily practice in order to monitor the population at risk. This is especially important because OCs are used by millions of women in the European Union (EU).[2,4] Because of the widespread use of OCs, many women may experience adverse events; in particular, cardiovascular and cerebrovascular adverse outcomes may affect women with additional risk factors (e.g., older age, obesity, diabetes mellitus, smoking). Although many studies have been performed in recent decades to unravel the contribution of these risk factors, data on the number of women currently using OCs are limited, let alone data on those at increased risk of developing adverse effects.

This study aimed to describe OC use in Europe by assessing the prevalence of use, user characteristics and prescribing patterns in daily practice in areas within three different EU countries. The results of this study are meant to provide a basis for risk minimisation planning.

Methods

Data sources

The user and treatment characteristics of OCs were studied in a retrospective multi-database study in 2009 and 2010. To capture patterns of use across Europe, the study included the PHARMO Database Network and Integrated Primary Care Information Project (IPCI) from the Netherlands, the Health Improvement Network (THIN) database from the UK and the Health Search Database (HSD) from Italy. These databases are part of the EU-ADR (adverse drug reactions) alliance network. Other databases in the network were not able to capture OC use due to the limited reimbursement (not captured

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in claims/reimbursement-based databases) or lack of primary care data (prescriptions regardless of reimbursement).

The PHARMO Database Network includes several linked databases which contain drug-dispensing and hospital records of about 4 million individuals in geographically defined areas in the Netherlands. This is about 23% of the population of the Netherlands. The database dates back to 1998.[5] The IPCI contains data from computer-based records of 1.2 million patients registered with a group of general practitioners (GPs) in the Netherlands, who voluntarily chose to supply data to the database. Some data are available from 1992; however, most are from after 2007, when a large number of practices were added.[6] Both the PHARMO and IPCI populations resemble the Dutch population in terms of age and sex distribution. In addition, the pharmacies (PHARMO) and participating GPs (IPCI) are spread throughout the Netherlands.[6,7] THIN is a database of primary care medical records of 2.7 million patients located throughout all regions of the UK who are registered with GPs who use the Vision GP computer system (InPractice Systems, London, UK). The database dates back to 1985.[8,9] The HSD contains data from computer-based patient records of more than 650 GPs in Italy (covering a total of 1.2 million active patients aged 15+, which is about 2.3% of the Italian population) who voluntarily agreed to collect data for the database and attended specific training courses for data entry; all use the same GP computer system. These GPs have been shown to be similar to the general population of GPs in terms of compliance with standards of care for chronic diseases.[10] The database dates back to 1998.

In each country (the Netherlands, UK and Italy), OCs are predominantly prescribed by GPs and dispensed through community pharmacies. A prescription is mandatory in all three countries, but in the Netherlands and Italy (not the UK) women may go directly to the pharmacy for resupply (in Italy for up to 6 months) and the prescription does not need to be renewed each time it is dispensed. Because the initial prescription is mandatory, all users in IPCI, THIN and HSD who were prescribed an OC by their GP, and all users for whom the GP reissues the prescription, are captured in the databases. PHARMO captures all community-dispensed OCs (primary and secondary care). Completeness of exposure data in IPCI and THIN depends on the extent to which the GP records capture all pharmacy dispensing records, for example when the same system is used. For HSD, completeness in recording of the first prescription every 6 months is expected to be high (because GPs are trained to record their prescriptions); however, as mentioned before, all resupplies in the following 6 months may not be recorded. For all databases, OC prescriptions are recorded irrespective of reimbursement requirements. In IPCI and THIN, GPs are not allowed to keep a paper-based patient record and are only allowed to provide data to the database in case minimal criteria for data quality are met. In addition, both in the Netherlands and in the UK, incentives were recently initiated which improved coding. IPCI and THIN do not directly link to hospital data. Both databases, however, capture information on referral and discharge letters but not necessarily from all practices. Information bias cannot be excluded; however, it is unlikely that it would differ between OC users and non-users.

Study population

The source population included all women in the databases at any time in the years 2009 to 2010. The study population was limited to women of reproductive age, defined as 15 to 49 years at index date, according to the World Health Organization definition.[4]

Study follow-up started after a run-in period of 1 year, beginning on 1 January 2009 (start of the study period) or at database entry whichever came later. Study follow-up ended at database exit (death, transferring out of the database or end of data collection) or at the end of the study period (31 December 2010), whichever came first. The study population includes users and non-users of OCs. From the study population, all women with an OC prescription (IPCI, THIN, HSD) or dispensed OC (PHARMO, hereafter also referred to as 'prescription') during the study period were selected as users. The index date for users was defined as the date of the first prescription during study follow-up. The index date for non-users was defined as the start date of the study follow-up.

Exposure definition

OCs in the study included all oral preparations listed under Anatomical Therapeutic Chemical (ATC) code G03AA (progestogens and oestrogens, fixed combinations), G03AB (progestogens and oestrogens, sequential preparations) and G03AC (progestogens) and corresponding British National Formulary (BNF)/Multilex codes for THIN (see ATC classification system [11]). The following chemical substances concern non-oral formulations that were not included: G03AA13 (norelgestromin and oestrogen transdermal preparation), G03AC03 (only excluded for PHARMO and IPCI, as oral levonorgestrel is not available for contraception in the Netherlands), G03AC06 (medroxyprogesterone acetate injection) and G03AC08 (etonogestrel subcutaneous implant). Users were classified as starters or existing users based on use of OC in the year preceding the index date (run-in period). Starters had a first prescription after at least one year of non-exposure. Note that the starter population includes first-time users ('starters') as well as re-starters after, for example, pregnancy. Existing users had had a previous OC prescription within the year before the index prescription. The proportion of women ever exposed to OCs was assessed in the entire database history (between database entry and index date) of users and non-users.

Episodes of continuous OC use were constructed from consecutive prescriptions in a user's complete database history. Uninterrupted use was defined as no temporal gaps between consecutive prescriptions. Discontinuation of an OC was defined at the end of the last prescription in the database or at a gap between expiry of a prescription and resupply. A switch was defined as prescription of another OC (i.e., other chemical substance or preparation) after previous OC use during study follow-up. Between switches, gaps were allowed.

User characteristics

Demographic characteristics and health indicators were assessed on the index date among OC users and non-users for age-standardised comparison. Diagnoses of hypertension,

diabetes mellitus, respiratory disease, systemic lupus erythematosus, rheumatoid arthritis and multiple sclerosis were ascertained from the entire history of the IPCI, THIN and HSD databases. As these conditions do not lead to hospital admission, ascertainment was not made in the PHARMO database, which captured only hospital admissions. Use of drugs for chronic conditions (at least two prescriptions in the year preceding the index date of drugs used in diabetes, antihypertensives, thiazides, beta-blockers, calcium antagonists, angiotensin-converting enzyme inhibitors, angiotensin II antagonists, lipid-modifying agents or drugs for obstructive airway diseases) and diagnoses of conditions associated with the use of OCs (deep vein thrombosis, pulmonary embolism, cerebrovascular disorders, myocardial infarction, breast and cervical cancer) were identified in all databases. As the PHARMO database captures hospital admissions only, deep vein thrombosis and other conditions not leading to admission were not well captured. Other variables of interest were parity and smoking; however, the availability of this information was limited. Smoking status information was available in IPCI and THIN, and some parity information was available in THIN, but for less than 5% of the source population. Recording of smoking status and parity status by the GP is not systematically done and is likely to have been selective. These variables could not therefore be analysed.

Coding and statistical analysis

Health indicators and morbidity were assessed using different disease-coding terminologies, depending on the database: (i) International Classification of Diseases, 9th revision, Clinical Modification for HSD and PHARMO; (ii) International Classification of Primary Care for IPCI; and (iii) read codes for THIN. Exposure of interest was assessed by searching for specific ATC codes. For the THIN database, where coding of drugs is based on BNF codes, BNF codes were mapped into ATC codes. The process of mapping of event data extraction from the different databases was based on medical concepts derived from the Unified Medical Language System and was adopted from the process previously described in other publications.[12–14]

A distributed network approach was adopted for the database study using Jerboa. This extraction method was developed as part of the FP7 (Seventh Framework Programme for Research)-funded EU-ADR project. Jerboa is a JAVA-based software that can elaborate the databases locally and produce aggregated output datasets that are shared in a remote research environment for further analyses.[15]

A point prevalence of OC use, irrespective of whether users were starters, re-starters or prevalent users, was calculated on 1 January 2010. The (re-)start rate per 100 person-years was calculated as the number of (re-)starters during the study period (numerator) divided by the non-exposed person-years in the study population. Other statistics include frequency tables and median and interquartile ranges (IQR) of user characteristics and OC use characteristics. Proportions of health indicators per 1000 women were age-standardised among non-users by using the age distribution (5-year groups) of OC users.

Results

Among 2.16 million women aged 15 to 49 years, 16.0% (95% CI 16.0–16.1) used OCs on 1 January 2010. The prevalence ranged from 19.7% (95% CI 19.7–19.8) in PHARMO to 2.6% (95% CI 2.6–2.7) in HSD. During 2009 and 2010, mainly second-generation progestogens were prescribed in the Netherlands (79.4% and 78.3% of users), mainly second- and third-generation progestogens (57.9% and 43.6% of users, respectively) were prescribed in the UK, and mainly third-generation progestogens were prescribed in Italy (61.8%). Most switches were from lower generation to third- or fourth-generation pills. Proportions of chronic diseases among OC users tended to be higher than age-standardised proportions among non-users. Proportions of women with a history of diseases associated with use of OC were lower among users than among non-users of the same age.

Prevalence and (re-)start rate of OC use

The source population comprised 4.9 million women from four health care databases in the Netherlands, UK and Italy, of whom 2.4 million were between the ages of 15 and 49 (study population). In this age group, 629,848 women (26.6%; 95% CI 24.8–28.4) had at least one OC prescription during the study period (Table 1). In the Netherlands (PHARMO, IPCI) and UK (THIN), this figure was 24.2–29.3% in the study period; in Italy (HSD) it was 10.8%. The percentages of OC users among women ≤15 years, 50–54 years and ≥55 years of age were, respectively, 1.7%, 5.0% and 0.4% (data not shown).

On 1 January 2010, 345,779 of 2,156,385 (16.0%; 95% CI 16.0–16.1) women in the study population were using OCs. The prevalence of OC use ranged from 2.6% (95% CI 2.6–2.7) in HSD to 19.7% (95% CI 19.7–19.8) in PHARMO.

The overall (re-)start rate of OC use was 5.8 (95% CI 5.8–,5.8) per 100 person-years and ranged from 3.0 (95% CI

Table 1. Prevalence and (re-)start rate of OC use in 2009 to 2010 among women aged 15 to 49 years.

Variable	Total	PHARMO	IPCI	THIN	HSD
Women, *n** (%)	2369.2 (100.0)	932.6 (100.0)	176.6 (100.0)	977.8 (100.0)	282.1 (100.0)
OC users, *n** (%)	629.8 (26.6)	273.1 (29.3)	42.8 (24.2)	283.5 (29.0)	30.5 (10.8)
Prevalence					
Population size at 1 January 2010, *n**	2156.4	897.1	148.7	842.2	268.3
OC users at 1 January 2010, *n**	345.8	176.9	14.5	147.3	7.0
Prevalence per 100 women, % (95% CI)	16.0 (16.0–16.1)	19.7 (19.7–19.8)	9.7 (9.6–9.9)	17.5 (17.4–17.5)	2.6 (2.6–2.7)
(Re-)start rate					
Unexposed person-years, *n**	3270.9	1302.2	219.9	1257.9	490.9
OC starters, *n*****	190.0	69.0	21.6	84.5	14.8
Start rate per 100 person-years, *n* (95% CI)	5.8 (5.8–5.8)	5.3 (5.3–5.3)	9.8 (9.7–10.0)	6.7 (6.7–6.8)	3.0 (3.0–3.1)

*Numbers are expressed in thousands.
**No use in year before first prescription in 2009 to 2010.

3.0–3.1) per 100 person-years in HSD to 9.8 (95% CI 9.7–10.0) per 100 person-years in IPCI.

User characteristics

The largest user age groups were 15 to 19 years in the Netherlands (19.2% of users in PHARMO and 21.6% of users in IPCI) and 20 to 24 years in the UK (21.5% of users) and Italy (17.3% of users) (Table 2). The median age of OC users was 25 to 29 years in the Netherlands and 30 to 34 years in the UK and Italy. The majority of users were prevalent, i.e., were already using OCs when the study period started. Re-starters had in general more database history before the index date, partly by definition (classification of users into either starter or re-starter largely depended on the available database history) and partly because starters are younger than re-starters.

The number of accumulated person-months of OC exposure before the study period was highest in PHARMO (median 28, IQR 11–55) and THIN (median 17, IQR 6–39), reflecting mainly a longer database history compared with IPCI and more OC use compared with HSD.

For all health indicators, except respiratory disease, proportions were low (Table 3). Although formal statistical testing was outside the scope of this descriptive study, proportions of chronic diseases among OC users tended to be higher than age-standardised proportions among non-users. Proportions of women with a history of deep vein thrombosis, pulmonary embolism, cerebrovascular disease, myocardial infarction, breast cancer or cervical cancer (all associated with OC use) were lower among users than among non-users of the same age. Either women stopped OCs upon occurrence of such an event (former users), or women had an event and therefore never started OCs (never users). The timing of OC use relative to the occurrence of these events was not studied.

Prescribing patterns

The types of OCs prescribed in 2009 and 2010 were assessed according to ATC code or prescription; thus, if a woman was prescribed two different OCs during the study period she was counted twice. In all databases, prescription of fixed combination OCs was most prevalent (Table 4). In the Netherlands (PHARMO and IPCI), levonorgestrel-containing preparations dominated this OC class with more than 70% of users. Second most prevalent were desogestrel- (prescribed to 10.0% of PHARMO users and 9.0% of IPCI users) and drospirenone-containing preparations (8.0% and 9.2%, respectively). In THIN, levonorgestrel-containing preparations were also most prevalent and prescribed to 42.4% of users, followed by norgestimate-containing preparations and drospirenone-containing preparations, prescribed to 11.2% and 10.9% of users, respectively. In HSD, most women were prescribed gestodene-containing preparations (39.2%) or drospirenone-containing preparations (32.1%).

Sequential combinations, mainly gestodene-containing preparations, were most prevalent in HSD (14.7%). In the Netherlands and THIN, sequential combinations were used by less than 7% and were most often levonorgestrel-containing preparations.

Progestogen-only preparations, mainly desogestrel, were much more frequently used in THIN (30.8% of users) than in the other databases (up to 5.8% of users).

When classifying OCs by progestogen generation, mainly second-generation progestogens (levonorgestrel) were prescribed in the Netherlands (79.4% of prescriptions in PHARMO and 78.3% in IPCI). In the UK, second-generation progestogens were also those most frequently prescribed (57.9%), but third-generation progestogens (norgestimate) constituted 43.6% of prescriptions. In Italy, third-generation progestogens (gestodene) were the most frequently prescribed (61.8%).

During 2009 and 2010, up to 11.6% of users in PHARMO, IPCI and HSD either discontinued OC use or switched to another type (Table 5). In THIN, 18.7% changed OC use. These changes were primarily switches. Only in HSD were discontinuations as frequent as switches.

Most switches occurred between fixed combination OCs (39.7–60.5%). Only in THIN were switches from fixed combination OCs to progestogen-only OCs as frequent (27.7%). Frequent specific switches were from levonorgestrel- to

Table 2. Characteristics of OC users aged 15 to 49 years in 2009 to 2010.

Characteristic	PHARMO $n = 273.1$*	IPCI $n = 42.8$*	THIN $n = 283.5$*	HSD $n = 30.4$*
Age (in years), *n** (%)				
15–19	52.3 (19.2)	9.2 (21.6)	50.4 (17.8)	2.6 (8.4)
20–24	51.3 (18.8)	7.7 (18.0)	61.1 (21.5)	5.3 (17.3)
25–29	42.8 (15.7)	7.0 (16.5)	59.2 (20.9)	5.1 (16.7)
30–34	35.3 (12.9)	5.8 (13.7)	44.8 (15.8)	5.0 (16.3)
35–39	34.9 (12.8)	5.2 (12.2)	34.9 (12.3)	4.9 (15.9)
40–44	31.2 (11.4)	4.4 (10.2)	21.9 (7.7)	4.6 (15.0)
45–49	25.5 (9.3)	3.3 (7.8)	11.2 (4.0)	3.1 (10.3)
User status. *n** (%)				
Starter	38.3 (14.0)	12.8 (30.0)	44.5 (15.7)	7.3 (23.9)
Re-starter	30.6 (11.2)	8.8 (20.5)	40.0 (14.1)	7.6 (24.8)
Prevalent user	204.2 (74.7)	21.1 (49.4)	199.0 (70.2)	15.6 (51.3)
Database history before the study period (months), median (IQR))				
Starters	83 (37–136)	27 (20–37)	105 (34–180)	100 (56–145)
Re-starters	122 (79–141)	32 (23–45)	132 (61–181)	128 (92–176)
Prevalent users	96 (54–132)	17 (12–28)	83 (23–174)	115 (79–164)
Accumulated person-months of OC exposure before the study period, median (IQR)				
Re-starters	28 (11–55)	6 (3–18)	17 (6–39)	6 (2–18)
Prevalent users	44 (16–77)	11 (5–28)	29 (11–61)	17 (6–45)

User status at date of first prescription of OCs during 2009 to 2010. Starters, no previous OC prescription in database; re-starters, previous OC prescriptions in database but not in year preceding index date; prevalent users, OC prescriptions in year preceding index date.
*Numbers are expressed in thousands.

Table 3. Prevalence (per 1000 users/non-users) of specific health indicators and morbidity among OC users aged 15 to 49 and age-standardised prevalence among non-users.

	PHARMO		IPCI		THIN		HSD	
Health indicator	Users $n = 273.1$*	Non-users $n = 659.5$*	Users $n = 42.8$*	Non-users $n = 133.9$*	Users $n = 283.5$*	Non-users $n = 694.3$*	Users $n = 30.5$*	Non-users $n = 251.6$*
Diagnosis of chronic condition								
Hypertension	NA	NA	32.2	23.8	12.6	14.5	45.2	33.6
Diabetes mellitus	NA	NA	10.7	9.3	7.3	8.6	7.3	8.1
Asthma or COPD	NA	NA	114.5	85.6	164.3	139.7	66.2	51.3
SLE	NA	NA	5.7	3.7	0.9	1.2	0.5	0.9
Rheumatoid arthritis	NA	NA	7.4	6.5	3.7	3.4	2.4	2.2
MS	NA	NA	1.8	1.7	1.6	1.5	2.7	1.6
Use of drugs for chronic condition								
Antihypertensive drugs	0.8	0.3	1.1	0.5	0.4	0.4	1.0	1.0
Diuretics	4.9	2.3	6.2	4.6	2.3	3.2	4.6	3.1
Beta-blocking agents	11.0	4.0	20.0	12.0	6.0	6.0	12.0	9.0
Calcium channel blockers	2.3	1.0	2.3	1.1	1.6	2.4	2.2	1.9
Agents acting on the renin–angiotensin system	8.4	3.6	7.3	5.1	4.0	5.0	11.1	8.2
Lipid-modifying agents	5.2	2.6	4.9	3.7	2.6	3.7	2.4	2.6
Diabetes drugs	6.1	3.0	4.0	2.8	6.9	7.8	4.2	4.5
Drugs for obstructive airway diseases	43.0	17.0	49.0	34.0	66.3	54.2	43.8	28.2
Previous diagnosis of disease associated with use of OCs								
Deep vein thrombosis	0.1	0.1	5.1	7.1	3.1	4.2	6.1	6.1
Pulmonary embolism	0.5	0.9	1.6	3.1	1.0	1.9	0.2	0.3
Cerebrovascular disease	1.2	1.2	2.2	2.6	0.7	1.4	1.1	1.8
Myocardial infarction	0.5	0.3	0.8	0.6	0.1	0.2	<0.1	0.2
Breast cancer	0.6	1.6	1.1	3.5	0.3	1.2	0.9	3.9
Cervical cancer	0.2	0.3	1.3	1.5	0.2	0.5	1.0	0.7

User status at date of first prescription of OCs during 2009 to 2010. Diagnoses of chronic conditions were identified in the entire database history of the GP databases but not in PHARMO, which captures only hospital admissions and is insensitive for the selected conditions. Also, the hospital admission records capture only part of the 'disease associated with use of OCs'. 'Use' was defined as at least two prescriptions in the year preceding the index date. OC-related disease was identified in the entire available history of all four databases.

*Numbers are expressed in thousands.

COPD, chronic obstructive pulmonary disease; MS, multiple sclerosis; NA, not applicable; SLE, systemic lupus erythematosus.

Table 4. OC preparations prescribed to women aged 15 to 49 years in 2009 to 2010.

OC preparation	PHARMO $n = 273.1$*	IPCI $n = 42.8$*	THIN $n = 283.5$*	HSD $n = 30.4$*
Fixed combination preparation, n* (%)	254.6 (93.2)	39.9 (93.3)	210.6 (74.3)	25.2 (82.8)
Lynestrenol and oestrogen	4.2 (1.5)	0.6 (1.4)	–	–
Norethisterone and oestrogen	2.7 (1.0)	0.4 (1.0)	16.3 (5.7)	–
Levonorgestrel and oestrogen	195.3 (71.5)	30.8 (72.1)	120.3 (42.4)	2.1 (6.8)
Desogestrel and oestrogen	27.2 (10.0)	3.9 (9.0)	22.2 (7.8)	2.5 (8.2)
Gestodene and oestrogen	13.2 (4.8)	1.9 (4.5)	10.0 (3.5)	11.9 (39.2)
Norgestimate and oestrogen	1.4 (0.5)	0.3 (0.7)	31.7 (11.2)	–
Drospirenone and oestrogen	22.0 (8.0)	3.9 (9.2)	30.9 (10.9)	9.8 (32.1)
Sequential combination preparation, n* (%)	18.5 (6.8)	2.3 (5.3)	9.7 (3.4)	4.5 (14.7)
Levonorgestrel and oestrogen	17.7 (6.5)	2.1 (4.9)	7.1 (2.5)	–
Norethisterone and oestrogen	0.6 (0.2)	0.1 (0.2)	2.2 (0.8)	–
Desogestrel and oestrogen	0.02 (<0.1)	0.001 (<0.1)	–	0.8 (2.6)
Gestodene and oestrogen	0.1 (<0.1)	0.02 (0.1)	0.2 (0.1)	3.7 (12.1)
Dienogest and oestrogen	0.06 (<0.1)	0.1 (0.2)	0.2 (0.1)	–
Progestogen-only preparation, n* (%)	7.1 (2.6)	1.8 (4.3)	87.4 (30.8)	1.8 (5.8)
Norethisterone	–	–	23.4 (8.3)	–
Lynestrenol	0.001 (<0.1)	0.002 (<0.1)	–	–
Levonorgestrel	–	–	–	1.3 (4.4)
Desogestrel	7.1 (2.6)	1.8 (4.3)	67.5 (23.8)	0.4 (1.4)
By progestogen generation, n* (%)**				
II	216.7 (79.4)	33.4 (78.3)	164.0 (57.9)	3.3 (11.1)
III	48.3 (17.7)	7.8 (18.2)	123.7 (43.6)	18.8 (61.8)
IV	22.0 (8.1)	4.0 (9.4)	31.1 (11.0)	9.8 (32.1)

All preparations prescribed in 2009 to 2010 were included, i.e., a woman using two different preparations in the study period appears twice.

*Numbers are expressed in thousands.

**Generation II: lynestrenol, norethisterone and levonorgestrel; generation III: desogestrel, gestodene and norgestimate; generation IV: drospirenone and dienogest.

drospirenone-containing preparations in PHARMO (22.5%) and IPCI (20.5%) (second to fourth generation), from gestodene- to drospirenone-containing preparations in HSD (19.9%) (third to fourth generation) and from preparations containing levonorgestrel and oestrogen to desogestrel-only preparations in THIN (14.6%) (second to third generation).

Discussion

This descriptive study of OC use assessed the prevalence of use and the characteristics of users and OC treatments in four databases from the Netherlands, UK and Italy.

About 20 to 30% of women of reproductive age in the Dutch and UK databases were prescribed an OC during the

Table 5. Discontinuations and preparation switches among OC users aged 15 to 49 years in 2009 to 2010.

Discontinuation/switch	PHARMO $n = 273.1$*	IPCI $n = 42.8$*	THIN $n = 283.5$*	HSD $n = 30.4$*
No changes in OC use, *n** (%)	250.4 (91.7)	38.2 (89.4)	230.4 (81.3)	26.9 (88.4)
Switch or change, *n** (%)	22.6 (8.3)	4.5 (10.6)	53.1 (18.7)	3.5 (11.6)
≥1 switch	16.6 (6.1)	2.9 (6.8)	42.2 (14.9)	1.8 (5.8)
≥1 discontinuation	6.4 (2.3)	1.7 (4.0)	11.9 (4.2)	1.8 (6.0)
Total number of switches in 2009 to 2010	19.6 (100)	3.4 (100)	52.1 (100)	2.2 (100)
Type of switch, *n** (%)				
Within fixed combinations	11.7 (59.8)	2.0 (60.5)	20.7 (39.7)	1.1 (49.7)
Fixed to sequential combinations	1.7 (8.8)	0.3 (8.2)	1.4 (2.7)	0.3 (12.6)
Fixed combinations to progestogen-only	1.4 (7.2)	0.2 (6.1)	14.4 (27.7)	0.2 (7.8)
Within sequential combinations	0.07 (0.4)	0.01 (0.2)	0.1 (0.1)	0.005 (0.2)
Sequential to fixed combinations	1.9 (9.6)	0.2 (6.6)	1.0 (1.9)	0.3 (15.5)
Sequential combinations to progestogen-only	0.08 (0.4)	0.01 (0.3)	0.8 (1.5)	0.04 (1.8)
Within progestogen-only	–	–	5.2 (10.0)	0.002 (0.1)
Progestogen-only to fixed combinations	2.6 (13.1)	0.6 (17.5)	8.2 (15.7)	0.2 (11.2)
Progestogen-only to sequential combinations	0.1 (0.7)	0.02 (0.7)	0.3 (0.6)	0.02 (1.0)

*Numbers are expressed in thousands.

study period. The proportion of women ever exposed to OCs is likely to be much higher, as women switch between use and non-use because of pregnancy, delivery or other health-related reasons. The United Nation's World Contraceptive Use 2010 report, for example, cited a higher prevalence: 40% of Dutch women, 28% of UK women and 14% of Italian women used the pill. This was, however, a household survey limited to women who were 'married or in union'. A recent study reported that 19% of female adolescents received a prescription for hormonal contraception in 2011, mainly consisting of a progestogen and oestrogen fixed combination, which is in line with our data.[16]

Proportions of chronic diseases among OC users tended to be higher than age-standardised proportions among non-users. Formal statistical comparisons, however, were outside the scope of the study. Proportions of women with a history of disease associated with OC use tended to be lower among users than among non-users of the same age. Age standardisation was done by database because the completeness for the different conditions varied per database. For example, PHARMO only captured hospital admissions; therefore, not all conditions could be studied in this database. Any comparison should thus be between users and non-users, not between databases.

All proportions, except for respiratory disease, were low, which limited interpretation of the trends among OC users and non-users. Although completeness is not likely to depend on OC use, the likelihood of being diagnosed could be associated with OC use, as an OC prescription requires a GP visit. For the GP-based diagnoses, the higher disease proportions may well be the result of information bias.

The study was designed to provide an overall description of OC use in the countries included in the study. The data covered only oral preparations, as OCs were the most widely used form of contraception in the study countries. Along with the choice of describing data from the overall user population of three different countries came the limitation of not being able to study other objectives. For example, as the study included incident as well as prevalent users, the focus on new users, who are at the highest risk of adverse events, was lost. In addition, the study did not investigate whether the prescribed OC formulations varied depending on age or comorbidity. Both objectives can be studied in the databases but were beyond the scope of the study.

Each country and database included in the study demonstrated different patterns regarding the type of OCs prescribed. In the Netherlands, fixed combinations of oestrogen and levonorgestrel (a second-generation progestogen) were prescribed for the majority of women; these preparations are first choice in the Dutch guidelines on hormonal contraception.[17] In the UK, levonorgestrel-containing preparations were prescribed to many women, but there were also many users of fixed combinations of oestrogen with norgestimate (third-generation progestogen) or drospirenone (fourth or 'other' generation).[18,19] There was substantial use of progestogen-only pills in the UK (30.8% of total use), which is confirmed by the study by Cea-Soriano *et al.* that reported use of progestogen-only pills among 25.5% of all users.[20]

In Italy, women were mainly prescribed fixed combinations of oestrogen with gestodene (third-generation progestogen) or drospirenone, although the Italian consensus document on thrombotic risk [21] recommends second-generation OCs as a first choice. Switches were most frequently to fourth-generation pills in the Netherlands and Italy, and to third-generation progestogen-only pills in the UK. Third-generation and possibly also fourth-generation pills have been associated with a higher risk of venous thrombosis than second-generation progestogens.[22–25] These results suggest that more risk minimisation measures are warranted.

Finding a health care database that captures OC use is a challenge, as OCs are only partly reimbursed, the GP is not systematically involved in every single dispensed prescription, and women may obtain resupplies from internet pharmacies instead of community pharmacies, especially healthy women who do not need to visit a pharmacy for other reasons. European-wide monitoring of OC use and follow-up for complications through health care databases is attractive due to the large source population, but it is restricted because of limited reimbursement. A difference between the Dutch PHARMO and IPCI databases was apparent. The populations are similar but the underlying data are different. PHARMO has dispensing records and reported high prevalence and lower (re-)start rates, while IPCI has prescription records and reported lower prevalence and high (re-)start rates. The practice of allowing resupply on a previous prescription in the Netherlands (and Italy, see Methods) limits the sensitivity of the GP prescription data for actual OC use. In the UK, a prescription is also needed for

resupply, and UK results were similar to those of PHARMO. The proportions of women with a prescription at any time during the study period were more consistent (29% in PHARMO and 24% in IPCI) and indicate that although prevalence and start rate estimates differ, 'users' are identified similarly. Apart from different prescribing habits, the type of OCs observed in the databases is also influenced by local availability (less choice for prescribers), distribution channels (whether a preparation is captured by the database) and user preference for reimbursed preparations. For instance, in Italy, OCs in commerce are classified as *reimbursed* or *non-reimbursed* at a national level, and in administrative databases only reimbursed prescriptions are recorded. Because in 2011 the large majority of OC prescriptions were non-reimbursed,[26] Italian administrative databases cannot be used to perform studies on OC use.

Keeping in mind its limitations, this study provides the following conclusions. First, among OC users a somewhat higher prevalence of chronic diseases and less history of disease associated with OC use were observed; however, information bias cannot be ruled out. Second, second-generation pills were most frequently prescribed in the Netherlands and the UK; in Italy, third-generation pills were most prevalent. Preparation switches were often to a third- or fourth-generation OC.

Declaration of interest

This document expresses the opinion of the authors of the paper. It may not be understood to be or quoted as being made on behalf or of reflecting the position of the European Medicines Agency or one of its committees or working parties.

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